

EMCH 310 Examination #2

NAME: Solution

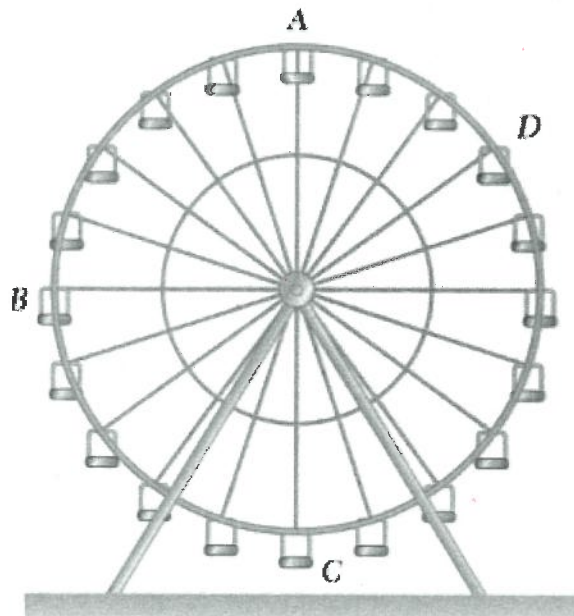
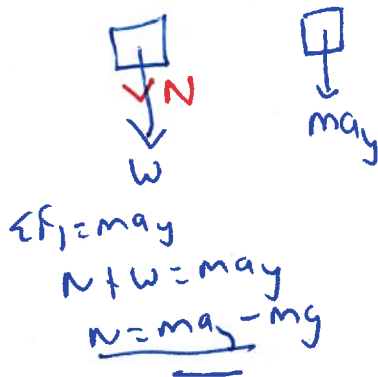
1) People sit on a Ferris wheel at Points "A, B, C and D.

The Ferris wheel travels at a constant velocity. At the instant shown, which person experiences the largest force from his or her chair?

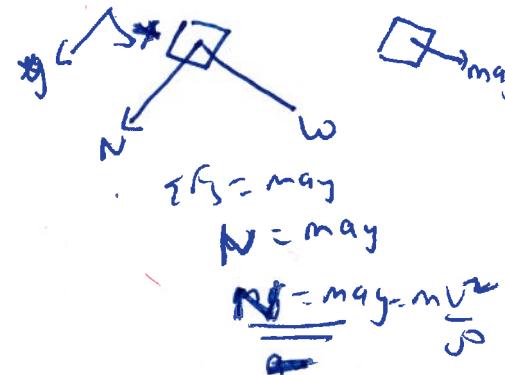
Assume you can neglect the size of the chair. Draw the FBD AND KD at each point and EXPLAIN YOUR FINAL ANSWER. NO CREDIT WILL BE GIVEN WITHOUT AN EXPLANATION AND A FBD AND KD AT EACH POINT.

- a) A
- b) B
- c) C
- d) D
- e) The force is the same for all passengers

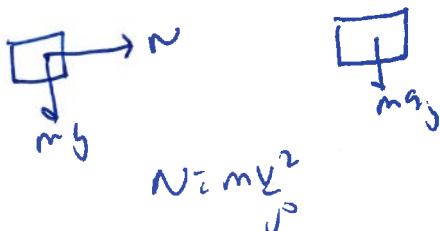
FBD and KD at A



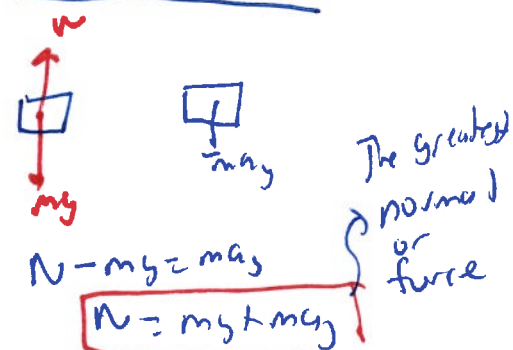
FBD and KD at D



FBD and KD at B



FBD and KD at C

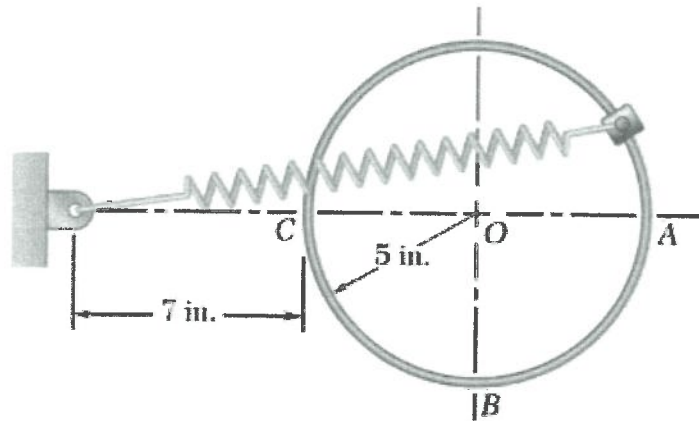


2) A collar of mass " m " is attached to a spring and slides without friction along a circular rod in a vertical plane.

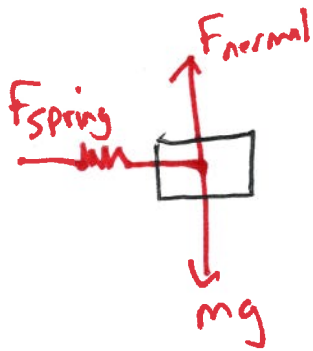
The spring has an undeformed length of 3 inches and a constant k .

The collar has a speed " v " at Point "C".

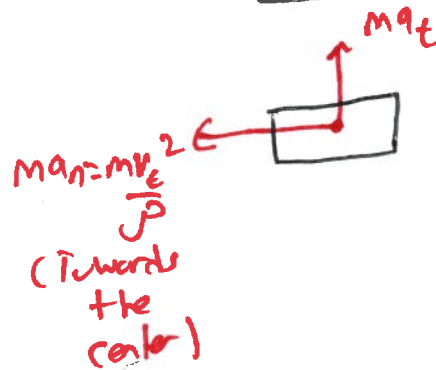
Draw the FBD and KD when the collar is at Point C.



FBD



KD

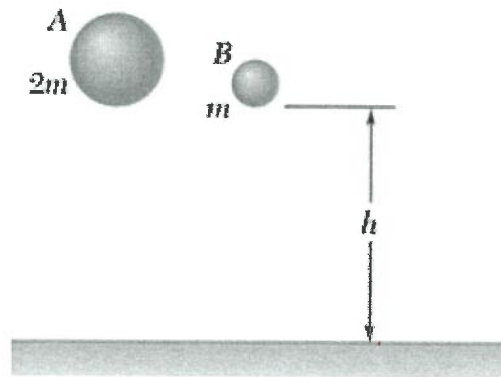


3) Two small balls A and B with masses $2m$ and m are released from rest at a height " h " above the ground.

Neglecting air resistance, which of the following statements are true when the two balls hit the ground?

SHOW YOUR ANSWER THROUGH EXPLANATION AND/OR CALCULATION

- a) The kinetic energy of A is the same as the kinetic energy of B
- b) The kinetic energy of A is half the kinetic energy of B
- c) The kinetic energy of A is twice the kinetic energy of B
- d) The kinetic energy of A is four times the kinetic energy of B



Each ball accelerates due to gravity at the same rate and so they have the same speed when they hit the ground.

$$T = \frac{1}{2}mv^2$$

Ball A

$$T = \frac{1}{2}(2m)v^2$$

$$T = mv^2$$

Ball B

$$T = \frac{1}{2}mv^2$$

$$T = 0.5mv^2$$

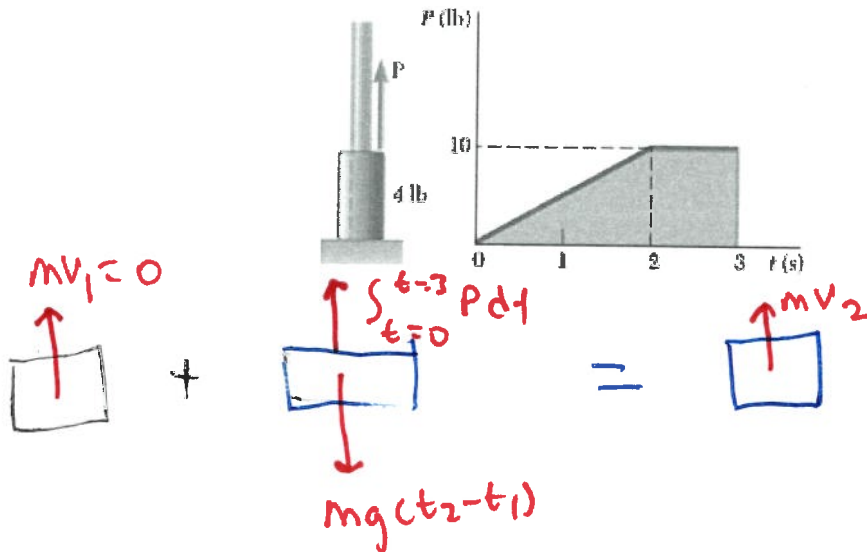
Ball A has twice the k.e of Ball B

(C)

4) A 4 lb collar which slides on a frictionless vertical rod is acted upon by a force "P" which varies in magnitude.

The collar is initially at rest.

Draw the impulse-momentum diagrams that could be used to determine the velocity.



5) What is a_A in m/s^2 ? SHOW ALL WORK

a) 2.09

b) 1.85

c) 1.63

d) 1.47

e) 1.22

$$a_A + 2a_B = 0$$

$$a_A = -2a_B \text{ so } a_B = -\frac{a_A}{2}$$

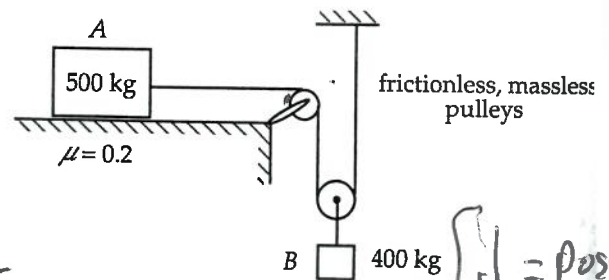
Body B

$$(400)(9.8) - 2T = 400 \frac{a_A}{2} \quad (1)$$

Body A

$$T - 0.2 \times 500 \times 9.8 = 500 a_A \quad (2)$$

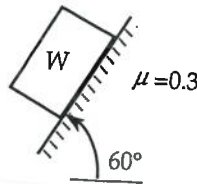
$$\text{From (1) and (2) } a_A = 1.63 \text{ m/s}^2$$



[down = positive]

6) How far, in meters, will the weight move in 10 seconds, if released from rest? SHOW ALL WORK

- a) 350
- b) 300
- c) 250
- d) 200
- e) 150



$$W \cos 60^\circ - W \mu \sin 60^\circ = \frac{W}{g} a$$

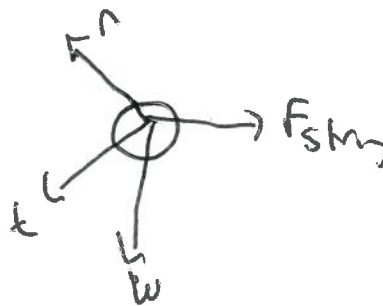
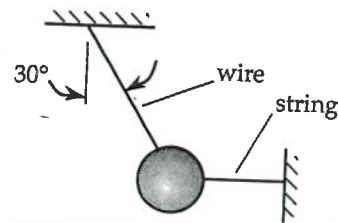
$$a = 9.8 (0.866 - 0.3 \times 0.5)$$

$$a = 7.02 \text{ m/s}^2$$

$$S = \frac{1}{2} a t^2 = 7.02 \times \frac{10^2}{2} = \underline{\underline{351 \text{ m}}}$$

7) Find the ratio of the tension in the wire before and immediately after the string is cut.

- a) $2/\sqrt{3}$
- b) $4/3$
- c) $2/3$
- d) $3/4$
- e) $\sqrt{3}/2$



Before

$$\sum F_y = 0$$

$$T \cos 30^\circ = W$$

$$T = \frac{W}{0.866}$$

After

$$F = 0$$

$$F_n = mg$$

normal is $\perp$ to tangent

$$T = W \cos \theta = 0.866 W$$

Ratio: $\frac{\text{Before}}{\text{After}} = \frac{1}{0.866 \times 0.866} = \underline{\underline{1.33}}$

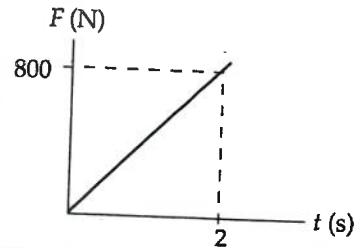
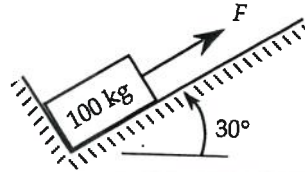
8) Using impulse-momentum, find the velocity, in m/s, of a 100 kg mass at $t=2$ seconds. The coefficient of friction is 0.2. SHOW ALL WORK

a. 0.245

b. 0.456

c. 0.567

d. 0.678



$$\int F dt = m \Delta v$$

$$F = 400t$$

$$F_f = fN = 0.2 \times 100 \times 0.866 \times 9.8 = 170$$

$$400t = 100 \times 0.5 \times 9.8 + 170$$

$$t = 1.655$$

$$\int_{1.65}^2 400t dt - (170 + 50 \times 9.8) \times 0.35 = 100v$$

$$v = 0.245 \text{ m/s}$$

9) If the coefficient of restitution is 0.8, find v_B' in m/s. SHOW ALL WORK

a) 16

b) 13

c) 11

d) 9

e) 7



$$10m - 20m = v_A' m + v_B' m$$

$$v_A' + v_B' = -10 \quad (1)$$

$$0.8 = \frac{v_B' - v_A'}{20 + 10}$$

$$v_B' - v_A' = 24 \quad (2)$$

From (1) and (2)

$$v_B' = 7 \text{ m/s}$$

$$v_A' = -17 \text{ m/s}$$

10) The 50 kg object strikes the unstretched spring. The maximum deflection is (SHOW ALL WORK)

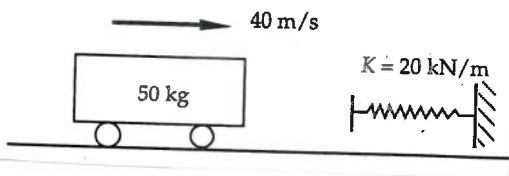
a) 1 m

b) 2 m

c) 3 m

d) 4 m

e) 5 m



$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

$$\frac{1}{2}(50)(40)^2 = \frac{1}{2}(20,000)x^2$$

$$x = 2 \text{ m}$$

Questions 11 and 12 refer to the following diagram.

A mass of 3 kg is initially at rest and is pulled along a frictionless, horizontal surface with a force shown as a function of time. (SHOW ALL WORK)

11) The acceleration of the block at $t = 2$ seconds is

a) $\frac{3}{4} \text{ m/s}^2$

b) $\frac{4}{3} \text{ m/s}^2$

c) 2 m/s^2

d) 8 m/s^2

e) 12 m/s^2

$t = 2 \text{ seconds}$

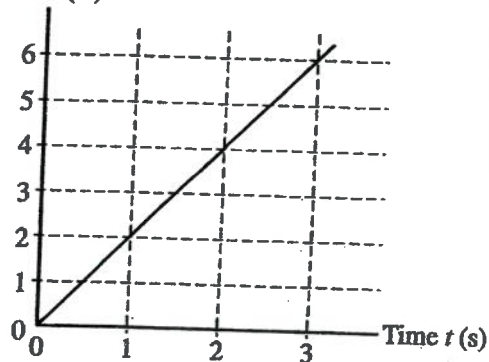
$F = 4 \text{ N}$

$F = ma$

$$a = \frac{F}{m} = \frac{4 \text{ N}}{3 \text{ kg}}$$

$$a = \frac{4}{3} \text{ m/s}^2$$

Force (N)



12) The speed of the block at $t = 2$ seconds is

a) $\frac{4}{3} \text{ m/s}$

b) $\frac{8}{3} \text{ m/s}$

c) 4 m/s

d) 8 m/s

$$m\vec{v}_1 + \sum \text{Impulse} = m\vec{v}_2$$

$$\sum \text{Impulse} = F \cdot t$$

$$\frac{1}{2}(2)(4)(2) = 3v_2$$

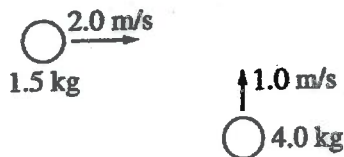
area time

$$8 = 3v_2$$

$$v_2 = \frac{8}{3} \text{ m/s}$$

Questions 13 and 14 refer to the following. SHOW WORK.

Two pucks moving on a frictionless air table are about to collide. The 1.5 kg puck is moving directly east at 2 m/s. The 4.0 kg puck is moving directly north at 1.0 m/s



13) What is the total kinetic energy of the two-puck system before the collision?

a) 5 J

b) 7 J

c) 10 J

d) 11 J

$$KE_x = \frac{1}{2}mv_x^2 = \frac{1}{2}(1.5)(2)^2 = 3J$$

$$KE_y = \frac{1}{2}mv_y^2 = \frac{1}{2}(4)(1)^2 = 2J$$

Kinetic energy is a SCALAR quantity

Direction does not matter

$$3J + 2J = \underline{5J}$$

14) What is the magnitude of the total momentum of the 2 puck system after the collision?

a) 1 kg m/s

b) 3.5 kg m/s

c) 5 kg m/s

d) 10 kg m/s

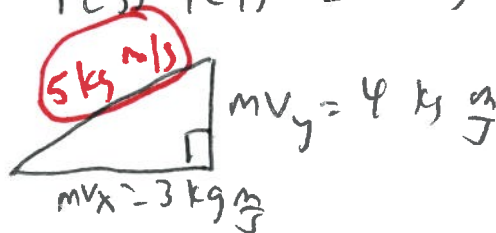
$$\text{Momentum}_x = mv = (1.5 \text{ kg})(2 \frac{m}{s}) = 3 \text{ kg} \frac{m}{s}$$

$$\text{Momentum}_y = mv = (4 \text{ kg})(1 \frac{m}{s}) = 4 \text{ kg} \frac{m}{s}$$

Momentum is a VECTOR quantity

Direction matters

$$\sqrt{(3)^2 + (4)^2} = 5 \text{ kg} \frac{m}{s}$$



Questions 15 and 16 refer to the following diagram. SHOW HOW YOU CAME UP WITH YOUR FINAL ANSWERS.

A system consists of 2 objects having masses m_1 and m_2 . ($m_1 < m_2$). The objects are connected by a massless string hung over a pulley as shown and then released.



$\downarrow = + \text{ (positive)}$

15) When the speed of each object is " v ", the magnitude of the total linear momentum of the system is

a) $(m_1 + m_2)v$

b) $(m_2 - m_1)v$

c) $\frac{1}{2}(m_1 + m_2)v$

d) $(m_2 v)$

The masses are different
Resultant momentum

$$\Delta p = m_2 v - m_1 v$$

$$\Delta p = (m_2 - m_1) v$$

16) When the object of mass m_2 has descended a distance " h ", the potential energy of the system has decreased by

a) $(m_2 gh)$

b) $\frac{1}{2}(m_1 + m_2)gh$

c) $(m_1 + m_2)gh$

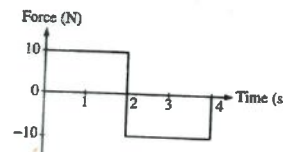
d) $(m_2 - m_1)gh$

One block's P.E goes
UP by $m_1 gh$
while the other block's
P.E goes down by
 $m_2 gh$

17) The graph below show the force on an object of mass M as a function to time.

For the interval 0 to 4 seconds, the total change in the momentum of the object is..... (SHOW HOW YOU CAME UP WITH YOUR FINAL ANSWER)

- a) 40 kg m/s
- b) 20 kg m/s
- c) -20 kg m/s
- d) 0 kg m/s

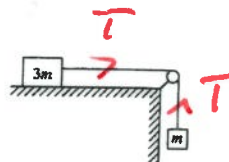


• Positive Impulse from 0 to 2 seconds
 $10 \text{ N} \times 2 \text{ s} = 20 \text{ N}\cdot\text{s}$

• Negative Impulse from 2 to 4 seconds
 $-10 \times 2 \text{ s} = -20 \text{ N}\cdot\text{s}$
 $20 - 20 = 0 \text{ kg m/s}$

18) A block of mass 3 kg can move without friction on a horizontal table. This block is attached to another block of mass m by a cord that passes over a frictionless pulley. If the masses of the cord and the pulley are negligible, what is the magnitude of the acceleration of the descending block? (SHOW WORK)

- a) 0
- b) $g/4$
- c) $g/3$
- d) $2g/3$



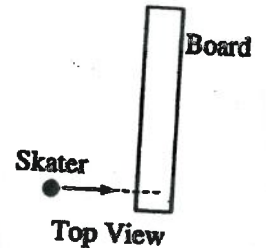
Horizontal
 Let $m = 3 \text{ kg}$
 $F_{\text{tension}} = 3ma \quad (1)$

Vertical
 $mg - F_{\text{tension}} = 0 \quad (2)$
 Substitute (1) into (2)
 $mg = 3ma$
 $a = \frac{g}{3}$

19) A long board is free to slide on a sheet of frictionless ice. As shown in the diagram, a skater skates to the board and hops onto one end causing the board to slide and rotate. Which of the following occurs? (CONCEPTUAL QUESTION-NO NEED TO EXPLAIN)

- a) Linear momentum is converted to angular momentum
- b) Kinetic energy is converted to angular momentum
- c) Translational kinetic energy is conserved

d) Linear momentum and angular momentum are both conserved



20) An object is thrown straight up. Which of the following is true about the sign of work done by the gravitational force while the object moves up and then moves down? (CONCEPTUAL QUESTION-NO NEED TO EXPLAIN)

- a) Work is positive on the way up, work is positive on the way down
- b) Work is negative on the way up, work is negative on the way down
- c) Work is positive on the way up, work is negative on the way down
- d) Work is negative on the way up, work is positive on the way down

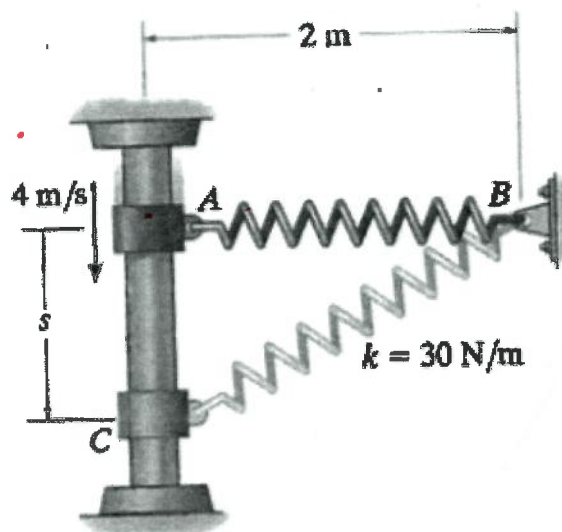
↓ = positive

21) A 5 kg collar has a downward velocity of 4 m/s at "A".

Determine the velocity of the collar when "s=2m".

The spring has an unstretched length of 1 m and a stiffness of $k = 25 \text{ N/m}$.

USE THE "CONSERVATION OF ENERGY" to solve.



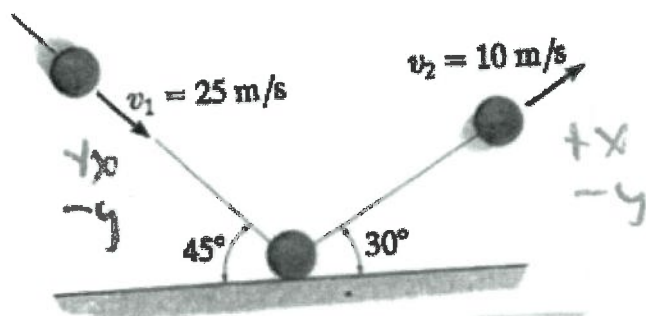
$$\begin{aligned} T_1 + \cancel{V_1} + V_{\text{elastic}} &= T_2 + V_2 + V_{\text{elastic}} \\ \text{energy}_1 & \quad \text{energy}_2 \\ \frac{1}{2}(5)(4)^2 + \frac{1}{2}(30)(2-1)^2 &= \frac{1}{2}(5)v^2 - (5)(9.81)(1) + \frac{1}{2}(25)(\sqrt{5}-1)^2 \\ 40 + 15 &= 2.5v^2 - 49.05 + 22.9 \\ 55 - 22.9 + 49.05 &= 2.5v^2 \\ 81.15 &= 2.5v^2 \end{aligned}$$

$$v^2 = 32.46$$

$$v = 5.69 \text{ m/s}$$

22) A 2 kg ball strikes the ground and then rebounds with the shown velocities. Assume the ball doesn't slip when it strikes the ground and neglect the size of the ball.

- Determine the magnitude of the impulse in the x-direction.
- Determine the magnitude for the impulse in the y-direction.
- Determine the resultant impulse.



x-direction

$$mv_{1x} + \int_{t_1}^{t_2} F_x dt = mv_{2x}$$

$$J_1 = \text{impulse in x-direction} = mv_{2x} - mv_{1x}$$

$$J_1 = (2)(10 \cos 30^\circ) - (2)(25 \cos 45^\circ)$$

a) $J_1 = \boxed{-18.1 \text{ N}\cdot\text{s}}$

y-direction

$$mv_{1y} + \int_{t_1}^{t_2} F_y dt = mv_{2y}$$

$$(-2)(25 \sin 45^\circ) + \int_{t_1}^{t_2} F_y dt = (2)(10 \sin 30^\circ)$$

$$J_2 = \text{impulse in y-direction} = 10 + 35.4$$

b) $J_2 = \boxed{45.4 \text{ N}\cdot\text{s}}$

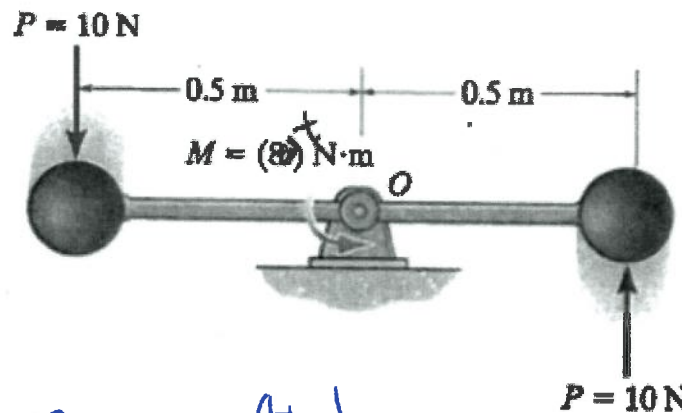
c) Resultant Impulse = $\sqrt{(-18.1)^2 + (45.4)^2} = \boxed{48.9 \text{ N}\cdot\text{s}}$

23) There are 2 identical 10-kg spheres shown and they are attached to a rigid rod that rotates about pin O.

Neglect the size of the spheres.

Tangential forces, $P=10$ N are applied to the spheres and the rod has a couple moment of $M = (80t)$ N·m, where t is in seconds.

What is the speed of the spheres when time = 4 seconds.



$$(H_2)_1 + \sum \int M_2 dt = (H_2)_2$$

$$+ \int_0^4 80t dt + 2 \left[\underbrace{(10)(0.5)}_{\text{moment}} (4) \right] = 2[(10)(0.5)v]$$

$$\frac{80t^2}{2} = 40t^2 \Big|_0^4 = 640 + 40 = 10v$$

$$680 = 10v$$

$$v = 68 \text{ m/s}$$